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NUMERICAL SIMULATION OF ONE-DIMENSIONAL TWO-PHASE FLOW USING A PRESSURE-BASED ALGORITHM

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Two-phase flow is studied numerically one-dimensionally using a two-fluid equation system. A hyperbolic type of governing equation is applied in our simulation. The segregated solution method is promoted and adopted in this article. By establishing mass conservation-based pressure correction, the SIMPLE algorithm is extended to two-phase flow simulations. Numerical results for well-known two-fluid air/water flow benchmark problems are presented and discussed.

1. INTRODUCTION

Multiphase flows are important in a large range of industrial applications, such as in safety analysis of nuclear power plants [1], the oil and gas industry [2], and the chemical and processing industry [3]. In the oil and gas industry, a growing number of oil and gas fields are situated far from land and at great depths. This has increased the drive toward field developments based on sub-sea processing and multiphase flow transitions. Therefore, it is of great importance to be able to predict flows in greater detail.

The one-dimensional (1D) two-fluid model is often used to obtain a realistic prediction for the gas–liquid flow inside a pipeline [4]. In general, the ability to predict this depends on the availability of mathematical models and experimental correlations. Among several two-phase flow models, there are three fundamentally different formulations of the macroscopic field equations: namely the two-fluid model, the mixture model [5], and the drift model [6]. Here, we focus on the two-fluid model, for which it seems that many authors agree on its basic form [7]. However, due to the lack of certain physical properties as well as the presence of complex eigenvalues (loss of hyperbolicity), the derivation and inclusion of additional terms in the two-fluid model has been widely studied. In this work, we consider a common variant of the two-fluid model.

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NOMENCLATURE

u_k	speed of each phase, m/s
p_k	pressure of each phase, Pa
p_k^{i}	pressure at gas–liquid interface, Pa
P	total pressure, Pa
g	acceleration of gravity, m/s^2
l_{e}	slope limiter
θ	inclination angle, °
c_k	sound speed of each phase, m/s
α_k	void fraction
ρ_k	density of each phase, kg/m^3
λ	eigenvalue
ψ	generic scalar variable
θ_{e}	smoothness parameter

Subscripts

k =g, l	
g	gas phase
l	liquid phase
e, w	east and west CV faces, respectively
P	main grid node associated to main CVs
E, W	nearest east and west nodes associated, respectively
u, d, c	upstream, downstream, and current node value, respectively

Superscripts

i	interface
n	current time level
n+1	next time level
κ	iteration counter
0	initial value

The computational methods used for solving the two-fluid model have been investigated by many researchers. Most research has focused on the coupled method for solving the model. In respect of first-order methods, several methods have been proposed such as the semi-implicit [8], the Stability enhancing two-step (SETS) method [9], and the implicit [5]. However, these methods have either large numerical diffusion or are not sufficiently stable [10]. In the 1990s and 2000s, various upwind schemes have been proposed for solving two-phase flow models—both mixture and two-fluid. Many of these schemes, often categorized as flux difference splitting (FDS) schemes, are based on suitable modifications of classical upwind schemes such as the Roe type [11]. Such schemes are accurate and robust, but they tend to be time-consuming due to the need for repeated calculation of the Jacobian of the system with respect to the conservative variables. Examples of such upwind schemes for two-phase flow models include implementations of the Roe scheme by Toumi et al. [12], Romate [13], Tiselj and Petelin [14], and Fjelde and Karlsen [15]. A rough Godunov scheme was implemented by Masella et al. [16]. For the approximation of a general two-fluid model, Coquel et al. [7] studied a kinetic upwind scheme that did not make use of the eigenstructure. Niu [17] explored hybrid flux-type flux-splitting schemes for a multicomponent flow model. An extension of the Advection upwind splitting (AUSM) method+ scheme for Euler equations was investigated by Paillère et al. [18] for this model. Evje and Fjelde [19] based the hybrid flux vector splitting (FVS)/FDS schemes on a crude approximation of the eigenstructure of the model. Yoem and Chang [20] proposed a modified HLLC-type Riemann solver for a compressible six-equation two-fluid model. García-Cascales and Paillère [21] applied AUSM-type schemes on a compressible two-phase flow problem. Chunge et al. [22] treated two-dimensional (2D) problems using the flux vector-splitting method. Yoem and Chang [23] simulated 2D two-phase flow using an approximate Jacobian matrix for the HLL (the approximate Reiman solver proposed by Harten, lax and van Leer) scheme.

Apart from coupled methods, segregated pressure-based methods also have the ability to deal with two-phase flow problems but compared with those mentioned

above, few published studies relate to segregated methods for two-phase flow. Darwish [24] proposed a class of pressure-based algorithm which focused on the multidimensional two-phase flow. Morales-Ruiz [25] focused on refrigeration systems with a 1D two-fluid model, but they adopted an ill-posed form of the two-fluid model. Issa and Kempf [26] simulated the transition from stratified to slug flow using a viscous two-fluid model employing the donor-cell difference scheme, but they focused on instability and capacity of the model and introduced simply their algorithm.

It is obvious that the segregated class of algorithm has not been fully developed and applied in 1D two-phase flow simulation. As a key branch of CFD technology, the segregated methods applied in compressible and incompressible flow simulation are generally accepted as efficient and practicable in engineering practice, and the merits of this class of algorithm should be carried forward to two-phase flow simulation. Therefore, in this article, based on a hyperbolic 1D two-fluid model, we focus on the detailed 1D numerical simulation of compressible two-phase flow by incorporation of the segregated method.

2. MODEL FORMULATION

The model with which we are concerned is formulated by stating separate conservation equations for mass and momentum for the two fluids. We will assume an isentropic model and no energy equation will be taken into account [27]. The 1D, inviscid, isentropic multiphase flow is customarily described by the continuity equation

$$\frac{\partial}{\partial t}(\alpha_k \rho_k) + \frac{\partial}{\partial x}(\alpha_k \rho_k u_k) = 0 \tag{1}$$

and the momentum equation

$$\begin{aligned} &\frac{\partial}{\partial t}(\alpha_k \rho_k u_k) + \frac{\partial}{\partial x}\left(\alpha_k \rho_k u_k^2\right) = -\alpha_k \frac{\partial p_k}{\partial x} \\ &- \left(p_k - p_k^i\right)\frac{\partial \alpha_k}{\partial x} + \alpha_k \rho_k g \sin\theta \end{aligned} \tag{2}$$

where mass transfer, wall friction, interface friction, and other possible effects are neglected [28]. In practical applications, one or more of these effects can be important. In the numerical study presented here, on the other hand, we wish to focus on the mathematically essential parts of the two-fluid model, keeping the number of parameters low. Such a practice is common [7, 29] and facilitates a comparison of results to benchmarks.

To close the system, the basic relation is used:

$$\alpha_g + \alpha_l = 1 \tag{3}$$

Moreover, an expression is needed for the relation between pressures in the different phases [10]. Like $p_l - p_g = p_g^i$ in this work, we assume that the liquid pressure is equal to the gas pressure, $P = p_g = p_l$. In addition, appropriate thermodynamical submodels are specified as previously described.

2.1. Thermodynamic Submodels

For phase k, we assume the simplified linear thermodynamic relations

$$p_k = c_k^2\left(\rho_k - \rho_k^0\right) \tag{4}$$

The compressibilities are constant, given by

$$\frac{\partial p_k}{\partial \rho_k} = c_k^2 \tag{5}$$

Throughout this work, for the liquid phase, we use the parameters

$$\rho_l^0 = 1,000\text{kg/m}^3$$

$$c_l = 1,000\text{m/s}$$

For the gas phase, we set

$$\rho_g^0 = 0$$

$$c_g = \sqrt{10^5}\text{m/s}$$

2.2. Interface Pressure Modeling

For the interface pressure p_i, the simple model, which is also employed in the CATHARE (the French two-phase flow simulation code capable of simulating thermal and mechanical phenomena occurring in the primary and secondary circlets of Pressurized Water Reactors for a wide variety of accidental situations) code [30], is chosen:

$$p_k - p_k^i = \Delta p = \gamma \frac{\alpha_g \alpha_l \rho_g \rho_l}{\alpha_g \rho_l + \alpha_l \rho_g}\left(u_g - u_l\right)^2 \tag{6}$$

Because it is commonly cited [7, 17–19, 21], the CATHARE model is our default expression for the interfacial pressure difference and we take $\gamma = 1.2$, following Evje and Flatten [19], unless otherwise stated.

Note that Eq. (6) is based on mathematical considerations and has little physical justification; consistency with Eq. (6) is our main motivation for choosing the model. What needs to be kept in mind is that the numerical techniques outlined in this article are also suitable for use in more general pressure-correction models.

3. HYPERBOLIC ANALYSIS

3.1. Characteristic Form

Writing the governing equations in quasilinear form

$$A\frac{\partial U}{\partial t} + B\frac{\partial U}{\partial x} = C \tag{7}$$

the solution vector is $U = [\alpha_g \quad p \quad u_g \quad u_l]^T$, and A, B, and C are the coefficient matrix.

The system of governing equations can be transposed into more compact form by with A^{-1}, which yields

$$\frac{\partial U}{\partial t} + A^{-1}B\frac{\partial U}{\partial x} = A^{-1}C \tag{8}$$

$$\frac{\partial U}{\partial t} + G\frac{\partial U}{\partial x} = D \tag{9}$$

The coefficient matrix $G = A^{-1}B$ becomes

$$G = \begin{bmatrix} \frac{\xi}{\psi} & \frac{\alpha_l\alpha_g(u_g-u_l)}{c_g^2c_l^2\psi} & \frac{\alpha_l\alpha_g\rho_g}{c_l^2\psi} & -\frac{\alpha_g\alpha_l\rho_l}{c_g^2\psi} \\ \frac{\rho_l\rho_g(u_g-u_l)}{\psi} & \frac{\zeta}{\psi} & \frac{\rho_l\alpha_g\rho_g}{\psi} & \frac{\rho_g\alpha_l\rho_l}{\psi} \\ \frac{\Delta p_{ig}}{\alpha_g\rho_g} & \frac{1}{\rho_g} & u_g & 0 \\ \frac{-\Delta p_{il}}{\alpha_l\rho_l} & \frac{1}{\rho_l} & 0 & u_l \end{bmatrix} \tag{10}$$

with

$$\xi = \frac{\alpha_l\rho_g u_g}{c_l^2} + \frac{\alpha_g\rho_l u_l}{c_g^2}$$

$$\psi = \frac{\alpha_g\rho_l u_g}{c_g^2} + \frac{\alpha_l\rho_g u_l}{c_l^2}$$

$$\zeta = \frac{\alpha_l\rho_g}{c_l^2} + \frac{\alpha_g\rho_l}{c_g^2}$$

$$\Delta p_{ik} = p_k - p_{ik}$$

and the source-term vector $D = [0 \quad 0 \quad g\sin\theta \quad g\sin\theta]^T$

3.2. Eigenvalues and Hyperbolicity

For the linearization of system around a given state U_0 to be mathematically well posed, a necessary condition is that the matrix $A(U_0)$ should be diagonalizable with real eigenvalues. The corresponding characteristic function is defined as $f(\lambda) = \det(G - \lambda I)$. An algebraic expression is obtained after applying $f(\lambda) = 0$. However, it is unfeasible to derive exact closed-form expressions for these. The approximate expressions derived by Evje and Flatten [19] used a perturbation technique suggested by Toumi and Kumbaro [12]. The system is testified to be hyperbolic. Here, the approximate eigenvalues of the coefficient matrix of the system are given. Further details can be found in Evje [19].

Introducing the perturbation parameter

$$\varepsilon = \frac{u_g - u_l}{\hat{c}_m\left(1 + \frac{\alpha_l \rho_g}{\alpha_g \rho_l}\right)} \tag{11}$$

where $\hat{c}_m$ is an approximate mixture speed of sound given by

$$\hat{c}_m = \sqrt{\frac{\alpha_g \rho_l + \alpha_l \rho_g}{(\partial \rho_g / \partial p)\alpha_g \rho_l + (\partial \rho_l / \partial p)\alpha_l \rho_g}} \tag{12}$$

It can be shown that the eigenvalues corresponding to pressure wave can be written as

$$\lambda_{1,2} = u_p \pm c_m \tag{13}$$

and that the eigenvalues corresponding to volume fraction waves can be expressed by

$$\lambda_{3,4} = u_u \pm \upsilon \tag{14}$$

Herein,

$$u_p = \frac{\alpha_g \rho_l u_g + \alpha_l \rho_g u_l}{\alpha_g \rho_l + \alpha_l \rho_g} + \hat{c}_m o(\varepsilon^3) \tag{15}$$

$$u_u = \frac{\alpha_g \rho_l u_l + \alpha_l \rho_g u_g}{\alpha_g \rho_l + \alpha_l \rho_g} + \hat{c}_m o(\varepsilon^3) \tag{16}$$

$$c_m = \hat{c}_m\left[1 + o(\varepsilon^2)\right] \tag{17}$$

and

$$\upsilon = \sqrt{\frac{\Delta p_i(\alpha_g \rho_l + \alpha_l \rho_g) - \alpha_g \alpha_l \rho_g \rho_l (u_g - u_l)^2}{(\alpha_g \rho_l + \alpha_l \rho_g)^2}} + \hat{c}_m o(\varepsilon^2) \tag{18}$$

Note that $\Delta p_i = 0$ renders υ imaginary and hence $\lambda_{3,4}$ complex. Therefore, one needs to take the factor $\gamma > 1$.

4. DISCRETIZATION

The governing Eq. (1) and (2) can be integrated in terms of the local averaged fluid variables using the finite control volume technique. The concept of a staggered mesh has been used, where velocities are defined at the cell boundaries, while pressure and void fraction are located at the middle of the main volume. A schematic representation of the control volume, together with the definition of the nodes, is depicted in Figure 1.

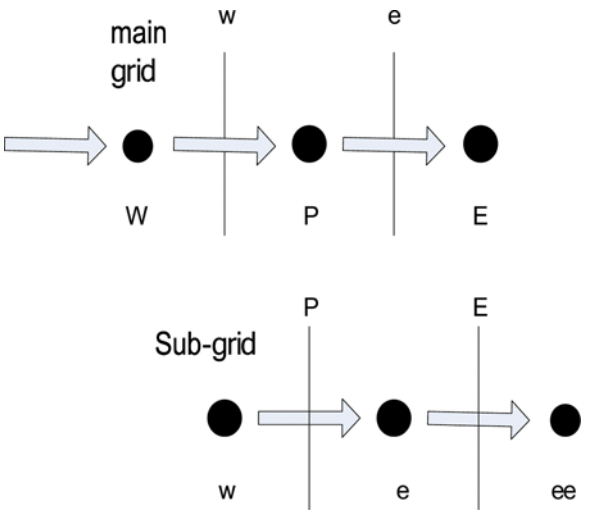


Figure 1. Staggered grid arrangement.

For momentum equations, subtracting the mass conservation equation multiplied by the velocity phase u_k, we have

$$(\alpha_k \rho_k)\frac{\partial u_k}{\partial t} + (\alpha_k \rho_k u_k)\frac{\partial u_k}{\partial x} = -\alpha_k \frac{\partial P}{\partial x} + \alpha_k \rho_k g \sin\theta + \Delta P \frac{\partial \alpha_k}{\partial x} \tag{19}$$

Discretizing the above equations in the sub-grid by the finite-volume method, the left-hand side then becomes

$$\mathrm{LHS} = (\alpha_k \rho_k)_{\mathrm{e}}^{n} \frac{u_{k\mathrm{e}}^{n+1} - u_{k\mathrm{e}}^{n}}{\Delta t} + (\alpha_k \rho_k u_k)_{\mathrm{e}}^{n} \frac{u_{k\mathrm{E}}^{n+1} - u_{k\mathrm{P}}^{n+1}}{\Delta x} \tag{20}$$

where LHS denotes the left-hand side of equation. The convection velocity terms $u_{k\mathrm{E}}$ and $u_{k\mathrm{P}}$ are evaluated as a function of the chosen numerical scheme. Main grid cell-face density and void fraction (ρ_{e}, α_{e}, respectively) must be calculated by interpolation. The interpolation method adopted here is the upwind type [31]:

$$(\alpha\rho)_{\mathrm{e}} = \left(\frac{1}{2} + \omega_{\mathrm{e}}\right)(\alpha\rho)_{\mathrm{P}} + \left(\frac{1}{2} - \omega_{\mathrm{e}}\right)(\alpha\rho)_{\mathrm{E}} \tag{21}$$

where the parameter ω value is either 0.5 (if $u_{\mathrm{e}} > 0$) or -0.5 (if $u_{\mathrm{e}} < 0$).

A common means of discretization is to use an upwind or donor-cell scheme [32]. However, in some cases, instabilities may be occur at large variable gradients (e.g., test case 3 in this study). Thus, the numerical scheme is employed and implemented within the context of the normalized variables and space formulation (NVSF) methodology [33]. Therefore, the value of the variable ψ at the face "e" is calculated by means of the expression

$$\psi_e = \left(1 - \frac{1}{2} l_e\right)\psi_P + \frac{1}{2} l_e \psi_E \quad (u_e \geq 0) \tag{22}$$

and

$$\psi_e = \frac{1}{2} l_e \psi_P + \left(1 - \frac{1}{2} l_e\right)\psi_E \quad (u_e < 0) \tag{23}$$

where l_e is a defined as slope limiter. For general schemes, the magnitude of l_e is decided on the basis of the direction of the flow and an estimate of the magnitude of local gradient, which is quantified by means of a smoothness parameter:

$$\theta_e = \frac{\psi_c - \psi_u}{\psi_d - \psi_c} = \frac{1}{\psi_E - \psi_P} \begin{cases} \psi_P - \psi_W & \text{for } u_e \geq 0 \\ \psi_{EE} - \psi_E & \text{for } u_e < 0 \end{cases} \tag{24}$$

Discussion in regard to the foregoing can be found in Sweby [34], Lenoard and Mokthtai [35], Thomas [36], and LeVeque [37], constraining the dependence of the limiter l on θ. If $(\psi_c - \psi_u)/(\psi_d - \psi_u)$ is less than 0 or greater than 1, there is a strong extremum between the nodes "d" and "u" and one takes $l_e = 0$ (i.e., the upwind rule). Several choices have been proposed for the monotone case in which $0 \leq (\psi_c - \psi_u)/(\psi_d - \psi_u) \leq 1$. Some popular limiters are discussed in detail in LeVeque [37].

Moreover, the integral value of the source term over the control volume is obtained by assuming the estimate of the source at the control volume center to represent the mean value over the whole control volume.

Furthermore, the additional term appearing in the momentum equations is treated explicitly, and their discretization is analogous to that of ordinary diffusion flux. So,

$$\Delta P \frac{\partial \alpha_k}{\partial x} \cong \Delta P_{ke}^n \frac{\left(\alpha_{kE}^n - \alpha_{kP}^n\right)}{\Delta x} \tag{25}$$

The pressure term is approximated as

$$\alpha_k \frac{\partial P}{\partial x} \cong \alpha_{k_e}^n \frac{P_E^{n+1} - P_P^{n+1}}{\Delta x} \tag{26}$$

Note that the explicit variables in the additional and pressure terms are handled by means of deferred correction [38] in the solution process.

Substituting face values by the functional relationships relating them to the neighboring node values, Eq. (17) is transformed after algebraic manipulation into the following discretized equation:

$$a_e(u_e)_k = a_w(u_w)_k + a_{ee}(u_{ee})_k + A_e(P_P - P_E) + b \tag{27}$$

where the coefficients a_w, a_e, and a_{ee} depend on the scheme selected and b is the source term of the discretized equation.

5. SOLUTION PROCEDURE

We replace the tie index $n+1$ by an iteration counter $\kappa+1$ and express the updated values of the variables at step $\kappa+1$ as those at the previous step plus an increment:

$$u_k^{\kappa+1} = u_k^{\kappa} + \delta u_k, \qquad P^{\kappa+1} = P^{\kappa} + \delta P$$

As mentioned in the original single-phase SIMPLE method [32], the neighboring-node increments are dropped, which corresponds to using the velocities u_k^{κ} at the previous iteration in evaluating the contribution of the neighboring nodes to the convective term of the momentum equation. With these relations, the equation becomes

$$\delta u_k = \left(\frac{A_{\mathrm{e}}}{a_{\mathrm{e}}}\right)_k (\delta P_{\mathrm{P}} - \delta P_{\mathrm{E}}) \tag{28}$$

5.1. The Pressure-Correction Equation

To derive the pressure-correction equation, the mass conservation equations of liquid and gas are added after dividing each by the representative values of the corresponding density, $\rho_{k,\mathrm{ref}}$. The additional one is then discretized in the main grid:

$$\begin{aligned}
&\frac{1}{\rho_{\mathrm{g,ref}}}\left[(\alpha\rho)_{\mathrm{ge}}^{\kappa} u_{\mathrm{g_e}}^{\kappa+1} - (\alpha\rho)_{\mathrm{gw}}^{\kappa} u_{\mathrm{g_w}}^{\kappa+1}\right] \\
&+\frac{1}{\rho_{\mathrm{l,ref}}}\left[(\alpha\rho)_{\mathrm{le}}^{\kappa} u_{\mathrm{l_e}}^{\kappa+1} - (\alpha\rho)_{\mathrm{lw}}^{\kappa} u_{\mathrm{l_w}}^{\kappa+1}\right] \\
&+\frac{\Delta x}{\Delta t}\left\{\frac{\left(\alpha_{\mathrm{g}}^{\kappa}\rho_{\mathrm{g}}^{\kappa+1}\right)_{\mathrm{P}} - (\alpha\rho)_{\mathrm{gP}}^{n}}{\rho_{\mathrm{g,ref}}} + \frac{\left(\alpha_l^{\kappa}\rho_l^{\kappa+1}\right)_{\mathrm{P}} - (\alpha_{\mathrm{l}}\rho_l)_{\mathrm{P}}^{n}}{\rho_{\mathrm{l,ref}}}\right\} = 0
\end{aligned} \tag{29}$$

Note that we do not pursue all variables at the same solution level; u_k and ρ_k are at the $\kappa+1$ level and α_k at the κ level. When we get convergence, the variables in $\kappa+1$ and κ are very close. This is a commonly used method of deferred correction.

To account for compressibility, the density increment is written as

$$(\rho_k)_{\mathrm{P}}^{\kappa+1} \cong (\rho_k)_{\mathrm{P}}^{\kappa} + \left(\frac{\partial\rho_k}{\partial P}\right)_{\mathrm{P}}^{\kappa} \delta P_{\mathrm{P}} = (\rho_k)_{\mathrm{P}}^{\kappa} + \left(\frac{1}{c_k^2}\right)_{\mathrm{P}}^{\kappa} \delta P \tag{30}$$

The gas and liquid void fractions are obtained from the continuity equations of each phase evaluated on the main mesh. The discretized equation is expressed as

$$\begin{aligned}
&(\alpha_k)_{\mathrm{P}}^{\kappa+1}(\rho_k)_{\mathrm{P}}^{\kappa+1} = (\alpha_k\rho_k)_{\mathrm{P}}^{n} \\
&+\frac{\Delta t}{\Delta x}\left[(\alpha_k\rho_k)_{\mathrm{e}}^{\kappa+1} u_{k_{\mathrm{e}}}^{\kappa+1} - (\alpha_k\rho_k)_{\mathrm{w}}^{\kappa+1} u_{k_{\mathrm{w}}}^{\kappa+1}\right]
\end{aligned} \tag{31}$$

5.2. Sequence of Operations

The key operations, in order of execution, are as follows.

1. Estimate the pressure field.
2. Solve the momentum Eq. (27) for velocity.
3. Solve the press-correction Eq. (29) and correct u_k, P, and ρ_k.
4. Solve the phase continuity Eq. (31) for void fractions.
5. Treat the corrected pressure as a new estimated pressure, return to step 2, and repeat the whole procedure until a converged solution is obtained.

6. RESULTS

In order to illustrate the capabilities of our algorithm in different situations, different test cases were used to test the numerical procedure and the quality of the numerical results in comparison to analytical expressions was obtained from the technical literature.

The pressure convergence criterion is that the relative value of the norm, which is calculated from the allowance of continuity equation, is less than the permissible value:

$$\frac{\sqrt{\sum_{n=1}^{N}(b^{\mathrm{p}})^2}}{q_{\mathrm{m}}} < \varepsilon_{\mathrm{p}}$$

where b^{p} is the mass source term of Eq. (29), q_{m} is mass flow rate at the inlet, and ε_{p} is the convergence criterion. In the following case, the time step is 0.0005 s and the convergence criterion is 10^{-6}.

6.1. Water Faucet Case

The water faucet case is described in Ransom [39], and it has become a common test case for 1D two-fluid models. The problem is a vertical tube 12 m in length; a schematic is shown in Figure 2. The top has a fixed volumetric inflow rate of water at a velocity of $u_{\mathrm{l}}^0 = 10\mathrm{m/s}$ and a liquid volume fraction of $\alpha_{\mathrm{l}}^0 = 0.8$, the bottom of the tube is open to ambient pressure, $P = 1 \times 10^5\mathrm{Pa}$, and the top of the tube is closed to vapor flow. Initially, the flow is uniform throughout the computational domain and the initial conditions are equal to the inlet conditions. A thinning of the liquid jet will take place due to the effect of gravity. Coquel et al. [7] provided the solution for gas void fraction and liquid velocity:

$$\alpha_{\mathrm{g}}(x,t) = \begin{cases} 1 - \dfrac{\alpha_{\mathrm{l}}^0 u_{\mathrm{l}}^0}{\sqrt{2gx + (u_{\mathrm{l}}^0)^2}} & if, x \le u_{\mathrm{l}}^0 t + \frac{1}{2}gt^2 \\ 1 - \alpha_{\mathrm{l}}^0 & \text{otherwise} \end{cases} \tag{32}$$

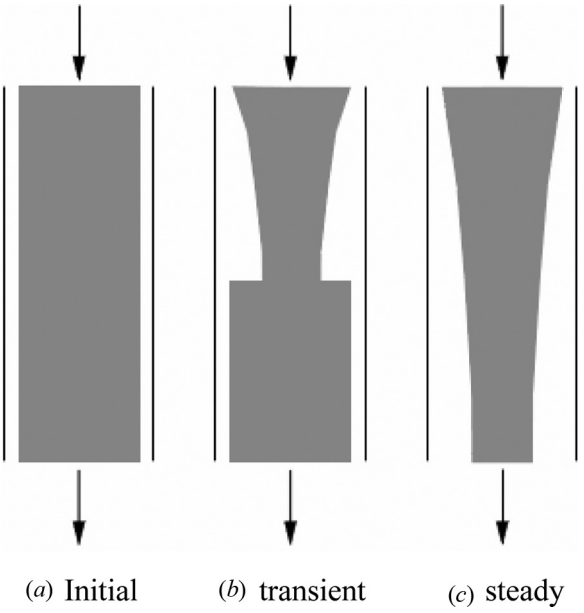


Figure 2. Water faucet case: (*a*) Initial, (*b*) transient, and (*c*) steady.

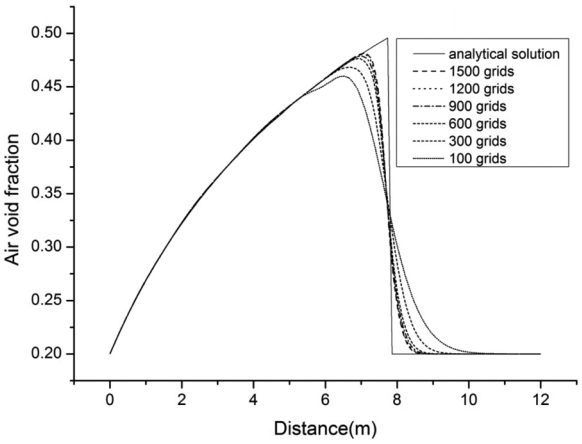
and

$$u_1(x,t) = \begin{cases} \sqrt{\left(u_1^0\right)^2 + 2gx} & \text{if, } x \le u_1^0 t + \frac{1}{2} g t^2 \\ u_1^0 + gt & \text{otherwise} \end{cases} \tag{33}$$

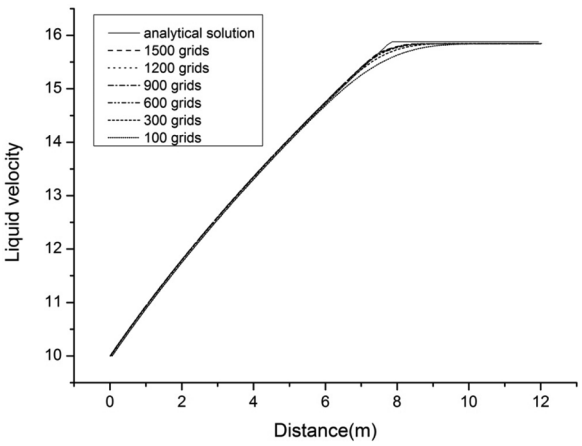
In this case, the pipe is assumed to be adiabatic and the fictional forces are neglected. The thermal effects are considered by our model, though it seems unnecessary to consider the thermodynamic in this air–water system. In order to test the convergence and the stability character of the algorithm, computations have been made using a discretization with 100 and 1,500 nodes. In this case, we must test different schemes, but it seems that there is no obvious influence from using different schemes. So, in this case, the results are calculated by using the first upwind scheme although the grid is somewhat dense.

Figure 3 shows a grid convergence study on void fraction and liquid velocity profile, at time $t = 0.6\,\text{s}$, computed by our model, together with a comparison to the analytical profiles. Since the grid number is higher, the discontinuity of the void wave is clearer. We chose the finest grid, $N = 1{,}200$, to prevent excessive numerical dissipation.

Figure 4 shows the time evolution of the void fraction and liquid velocity profile computed on mesh 1,200 nodes. In terms of profiles and as a grid convergence study, a close agreement with the analytic solutions is observed, though the dissipation of numerical results is not small for a scanty grid.



(*a*) Air void fraction



(*b*) Liquid velocity

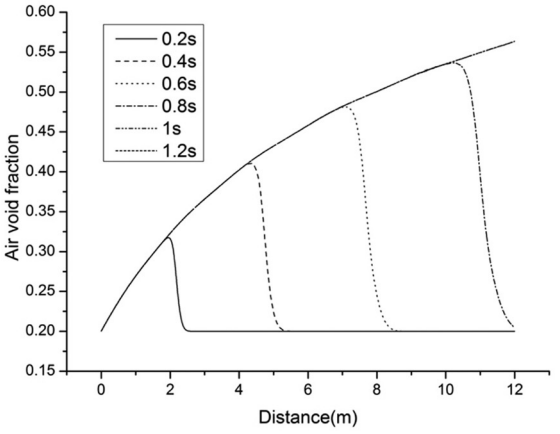
Figure 3. Water faucet problem (time $t = 0.6\,\mathrm{s}$); grid convergence study; (*a*) air void fraction and (*b*) liquid velocity.

6.2. Discontinuity Movement in the Uniform Flow Case

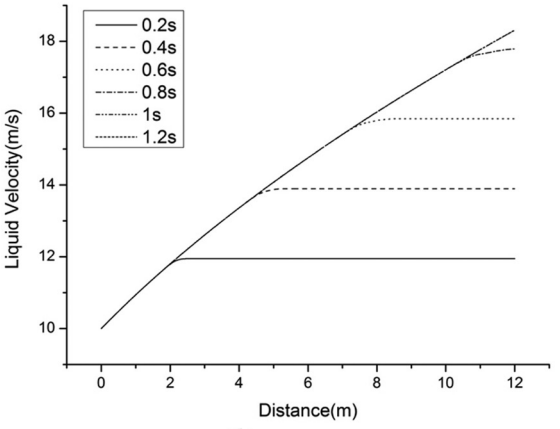
This case was proposed by Yeom and Chang [40]. The problem is a 1D horizontal pipe, 12 m long, with a diaphragm in its midsection and different left and the right states. The initial state is given by

$$\begin{pmatrix} \alpha_g \\ P \\ u_g \\ u_l \end{pmatrix}_L = \begin{pmatrix} 0.2 \\ 1 \times 10^6\,\mathrm{Pa} \\ 10\,\mathrm{m/s} \\ 10\,\mathrm{m/s} \end{pmatrix} \quad \begin{pmatrix} \alpha_g \\ P \\ u_g \\ u_l \end{pmatrix}_R = \begin{pmatrix} 0.8 \\ 1 \times 10^6\,Pa \\ 10\,m/s \\ 10\,m/s \end{pmatrix}$$

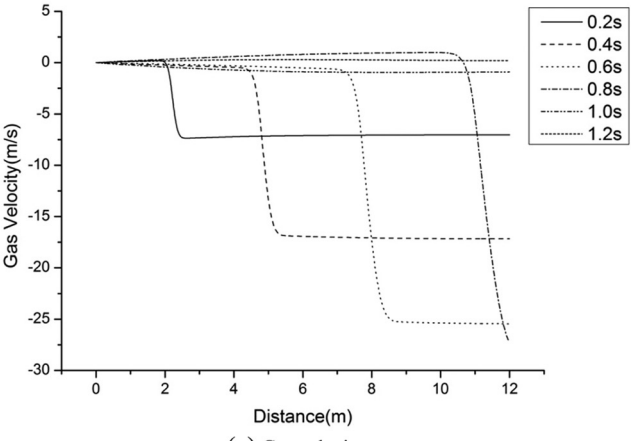
We consider a discontinuity movement in a uniform two-phase flow. In this case, the pipe is assumed to be adiabatic and the frictional forces are neglected.



(*a*) Air void fraction



(*b*) Liquid velocity



(*c*) Gas velocity

Figure 4. Water faucet problem—time evolution study: (*a*) air void fraction and (b) liquid velocity.

The air void fraction is advected along the computational domain [41] when the flow starts.

The solution is obtained using 100 grids with the first upwind scheme. The results are displayed at time $t = 0.2$ s. Figure 4 compares the exact solution to the numerical results. In this figure, pressure and velocities are uniform but the void fraction has discontinuity, which is advected to the right with some numerical dissipation. Although the first upwind scheme is adopted, the simulation results are still acceptable as in the previous case.

6.3. Sedimentation Case

This phase separation problem was proposed by Coquel et al. [7]. As shown in Figure 5, it consists of a vertical tube of length $L = 7.5$ m, closed at both ends. Initially it has a uniform pressure of $P = 1 \times 10^5$Pa and αl =0.5. At $t = 0$, the phases start to separate under the influence of gravity, which is the only source term considered. Approximate analytical solutions for liquid velocity and volume fraction, by assuming that the liquid is accelerated by gravity only, are given as

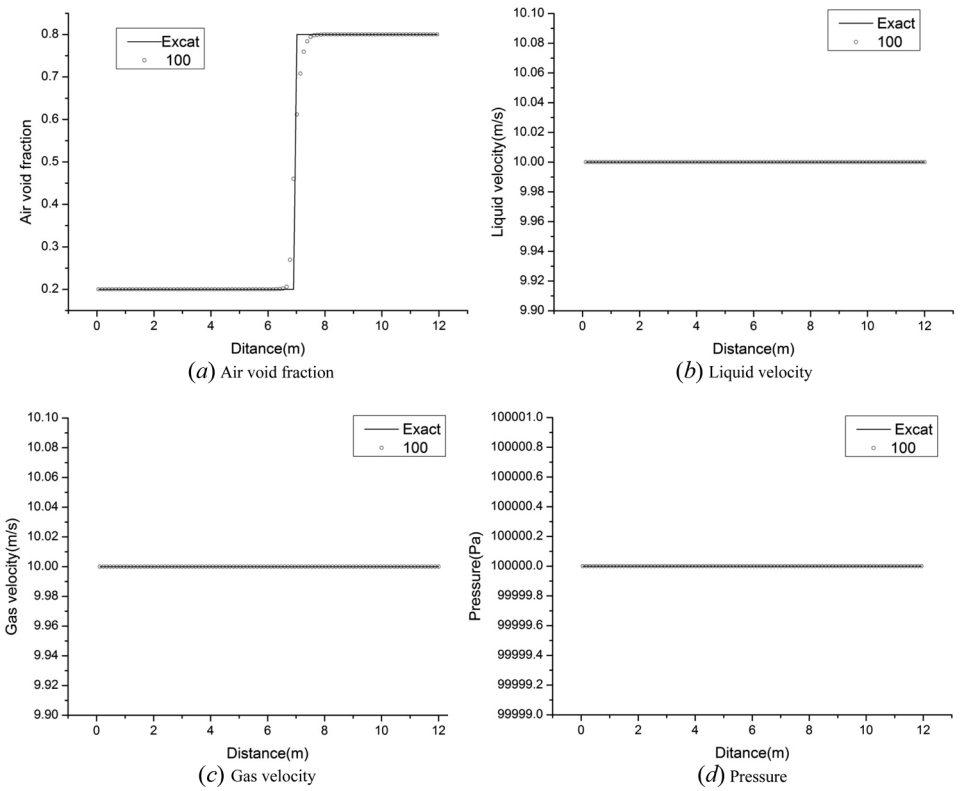


Figure 5. Discontinuity movement in a uniform two-phase flow: (*a*) air void fraction, (*b*) liquid velocity, (*c*) gas velocity, and (*d*) pressure.

$$\alpha_l(x,t) = \begin{cases} 0 & \text{for} \quad x < 1/2gt^2 \\ 0.5 & \text{for} \quad 1/2gt^2 \le x < L - 1/2gt^2 \\ 1 & \text{for} \quad L - 1/2gt^2 < x \end{cases} \tag{34}$$

and

$$u_l(x,t) = \begin{cases} \sqrt{2gx} & \text{for} \quad x < 1/2gt^2 \\ gt & \text{for} \quad 1/2gt^2 \le x < L - 1/2gt^2 \\ 0 & \text{for} \quad L - 1/2gt^2 < x \end{cases} \tag{35}$$

after time

$$T = \sqrt{\frac{L}{g}} \approx 0.87\text{s}$$

The phase should be fully separated in the idealized case, in which we compare the results for SMART [42] to the first upwind; the solution is obtained using 300 grids with both schemes. The results are displayed at time $t = 0.5$ s. Figure 6 shows a comparison between first upwind and SMART for the same grids. In this case, all variables except pressure have a large gradient, the first upwind scheme resulting in large numerical diffusion for simulation. However, high-order composite schemes have their advantages such as minor numerical diffusion and boundedness. In Figure 6, the SMART scheme is seen clearly to be more accurate, with no evidence of spurious oscillation or overshoot.

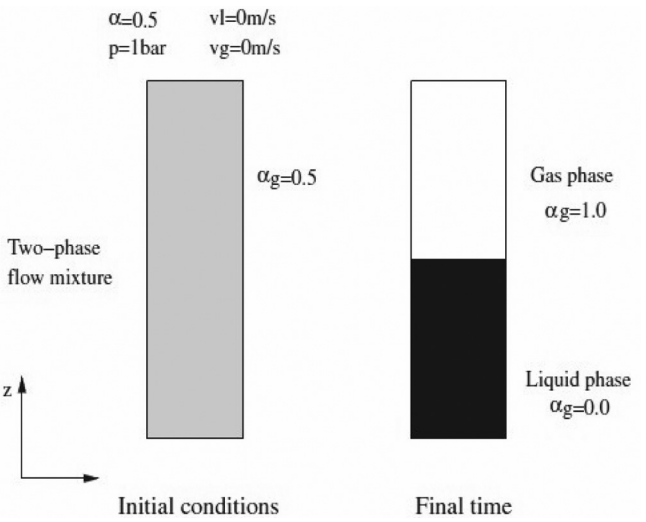
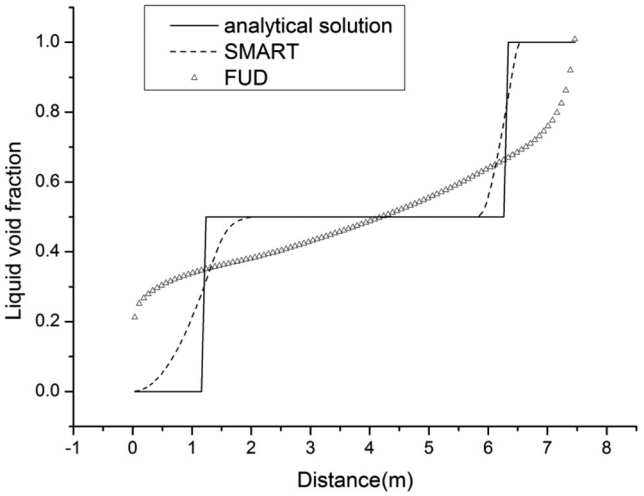
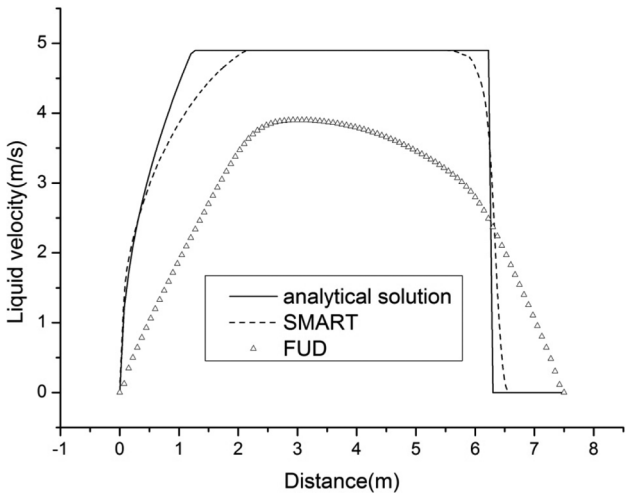


Figure 6. Sedimentation case.



(*a*) Liquid void fraction



(*b*) Liquid velocity

Figure 7. Sedimentation problems—comparison of FUD (first upwind) and SMART: (*a*) liquid void fraction and (*b*) liquid velocity.

7. CONCLUSION

In this article, 1D two-phase flow was studied numerically. Based on the hyperbolic two-fluid model, 1D two-phase flow was well simulated by the pressure-based method, which is an extension of the SIMPLE method. In fact, several ways of dealing with velocity increments, such as PISO (Pressure implicit split-oprerator) and SIMPLEC (SIMPLE consistent), are also available. Validation and verification of the algorithm are illustrated by some well-known benchmark cases. A good agreement between numerical results and analytical solutions, depending

on the case, was obtained with the model proposed in this article. In addition, taking case 3 for example, although the level of calculation for 1D simulation is generally well within the limits of most modern computers, high-order composite schemes for convection terms still play an important role in 1D simulation for dealing with spurious oscillation, overshoot, and false diffusion.

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